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4、边界条件：

$$P_{(x)}\left(y_1^*\frac{dy_2}{dx}-y_2\frac{dy_1^*}{dx}\right)\Big|_a^b=0$$

$\Rightarrow P_{(x)}\neq 0$ 时，

$$y_{1(b)}^{*}y_{2(b)}'-y_{2(b)}y_{1(b)}^{*'}=y_{1(a)}^{*}y_{2(a)}'-y_{2(a)}y_{1(a)}^{*'}$$

$$\left[\alpha_{11}^{*}y_{1(a)}^{*}+\alpha_{12}^{*}y_{1(a)}^{'*}\right]\left[\alpha_{21}y_{2(a)}+\alpha_{22}y_{2(a)}'\right]-\left[\alpha_{11}y_{2(a)}+\alpha_{12}y_{2(a)}'\right]\left[\alpha_{21}^{*}y_{1(a)}^{*}+\alpha_{22}^{*}y_{1(a)}^{*'}\right]=y_{1(a)}^{*}y_{2(a)}'-y_{2(a)}y_{1(a)}^{*'}$$

即：

$$\begin{cases}\alpha_{11}^{*}\alpha_{21}-\alpha_{11}\alpha_{21}^{*}=0\\ \alpha_{11}^{*}\alpha_{22}-\alpha_{12}\alpha_{21}^{*}=\dfrac{P_{(a)}}{P_{(b)}}\\ \alpha_{12}^{*}\alpha_{21}-\alpha_{11}\alpha_{22}^{*}=-\dfrac{P_{(a)}}{P_{(b)}}\\ \alpha_{12}^{*}\alpha_{22}-\alpha_{11}\alpha_{21}^{*}=0\end{cases}$$

5、(1)

$$\begin{cases}\nabla^2\Phi+\lambda\Phi=0\\ \Phi\big|_{\Sigma}=0\end{cases}$$

$\rightarrow\quad \Phi=\sum_{k=0}^{\infty}\lambda_k\Phi_k$

展开，$u(r,\theta,\varphi)=\sum_{k=0}^{\infty}C_k\Phi_{k(r,\theta,\varphi)}$，$f=\sum_{k=0}^{\infty}d_k\Phi_{k(r,\theta,\varphi)}\Rightarrow d_k=\dfrac{\int f\Phi_k\rho dv}{\int\Phi_k^2\rho dv}$

代入，得：$u(r,\theta,\varphi)=\sum_{k=0}^{\infty}\dfrac{\int f\Phi_k\rho dv'}{\lambda\int\Phi_k^2\rho dv'}\Phi_{k(r,\theta,\varphi)}$

(2)，此时，$f=-\dfrac{\rho(r,\theta,\varphi)}{\varepsilon_0}$，$\Phi_k=j_{l(k_{nl}r)}Y_l^m(\theta,\varphi)$

代入，$u=\sum_{n=1}^{\infty}\sum_{l=0}^{\infty}\sum_{m=-l}^{l}\dfrac{-2}{\varepsilon_0 k_{nl}^2\left[j_{n(l(k_{nl}))}\right]^2}\int_0^1\iiint j_{ll(k_{nl}r)}Y_l^{m*}(\theta,\varphi)\rho(r,\theta,\varphi)r^2\sin\theta dr d\theta d\varphi\; j_{ll(k_{nl}r)}Y_L^m(\theta,\varphi)$

(3)，此时，$\lambda_k=k_{mi}^2=\left(\dfrac{\mu_i^{(m)}}{a}\right)^2$，$\Phi=J_m(k_{mi}r)$

代入，$u(r,\theta,\varphi)=\sum_{m=0}^{\infty}\sum_{i=0}^{\infty}\dfrac{-2}{\varepsilon_0\mu_i^{(m)2}\left[J_n'(\mu_i^{(m)})\right]^2}\int_0^1\rho(r,\theta,\varphi)J_m\left(\dfrac{\mu_i^{(m)}}{a}r\right)rdr\;J_m\left(\dfrac{\mu_i^{(m)}}{a}r\right)$

P.9 1.

$$x(t) \rightleftharpoons X(t),\quad \dot{x}(t) \rightleftharpoons pX(t),\quad \ddot{x}(t) \rightleftharpoons p^2 X(t),\quad 1+\cos\omega t \rightleftharpoons 1+\frac{p}{p^2+\omega^2}$$

$$\Rightarrow\quad X(t) = \frac{P}{p^2+2\rho p+\omega_0^2}\,\frac{H}{p^2+\omega^2} = \frac{P}{p^2+\omega^2}\cdot\frac{\sqrt{\omega_0^2-\rho^2}}{(p+\rho)^2+\omega_0^2-\rho^2}\,\frac{H}{\sqrt{\omega_0^2-\rho^2}}$$

卷积:

$$x(t) = \frac{H}{\sqrt{\omega_0^2-\rho^2}}\int_0^t \cos\omega(t-\tau)\sin\left(\sqrt{\omega_0^2-\rho^2}\,\tau\right)d\tau$$